

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458904.7829 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.056 +/- 0.026
Transit depth (Rp/Rs)^2	0.31 +/- 0.29 [%]
Orbital Inclination (inc)	87.11 +/- 2.93
Airmass coefficient 1 (a1)	1.1559 +/- 0.0067
Airmass coefficient 2 (a2)	-0.104 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	4.74
Optimal Annulus	13.28
Transit Duration (day)	0.0528 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.056 ± 0.026 vs 0.1594 (deviation 3.69σ)
Duration fit vs published	0.0528 d vs 0.0512 d (deviation 0.3σ)
Mid-time O–C	-4.6 min
Depth SNR	2.0
Residual scatter	0.58 %

Light curve

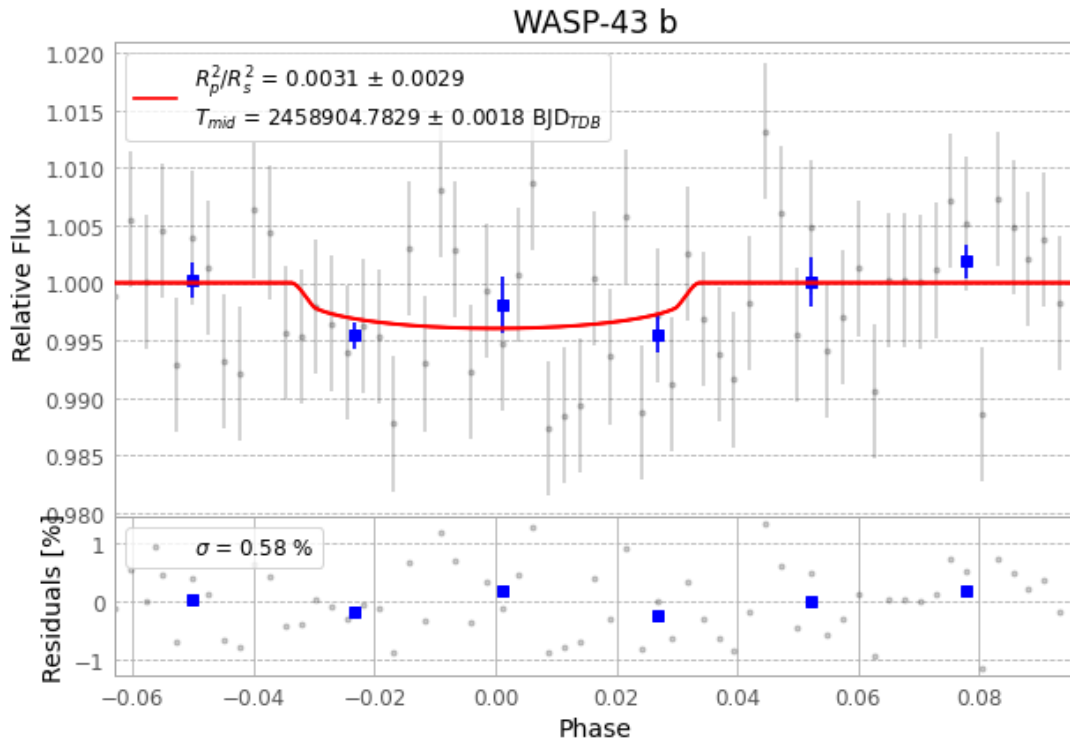


Figure 1 — FinalLightCurve_WASP-43 b_2020-02-24.png (SHA-256 0787c48a90043922...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 359], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2558d64dc2c18377...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), WASP-43200225052412.FITS → WASP-43200225083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200225.log (203dae2aa1ec...), FinalLightCurve_WASP-43 b_2020-02-24.png (0787c48a9004...), FinalLightCurve_WASP-43 b_2020-02-24.csv (c3e49854113f...)

Compute resources

Wall time 118.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:47Z.